String:-

In [ ]:
```
String is a collection characters surrounded by double quates and single quates,
Multiline string surrounded by triple signle quates nd triple double quates
ex.(''' Multiline String''')("""Multipple line string""")
```

In [2]:
```
string1="Python"
string2='Python'
string3='''Python'''
string4="""Python"""

print(string1)
print(string2)
print(string3)
print(string4)
string1
```

```
Python
Python
Python
Python
```

Out[2]:
```
'Python'
```

In [3]:
```
string1="Python and "data" science"
string1
```

```
  Cell In[3], line 1
    string1="Python and "data" science"
                        ^
SyntaxError: invalid syntax
```

string1='Python's 'Class' string1

In [4]:
```
string3="""Data science is an interdisciplinary field that combines statistics, machine
learning, programming (like Python and SQL), and domain knowledge to extract meaningful
insights from vast amounts of structured and unstructured data. It is used to drive
business decisions, optimize operations, and build predictive models"""
string3
```

Out[4]:
```
'Data science is an interdisciplinary field that combines statistics, machine \nlearni
ng, programming (like Python and SQL), and domain knowledge to extract meaningful\nins
ights from vast amounts of structured and unstructured data. It is used to drive \nbus
iness decisions, optimize operations, and build predictive models'
```

In [5]:
```
string4='''Data science is an interdisciplinary field that combines statistics,
machine learning, programming (like Python and SQL), and domain knowledge to
extract meaningful insights from vast amounts of structured and unstructured data.
It is used to drive business decisions, optimize operations,
and build predictive models'''
string4
```

Out[5]:
```
'Data science is an interdisciplinary field that combines statistics, \nmachine learni
ng, programming (like Python and SQL), and domain knowledge to \nextract meaningful in
sights from vast amounts of structured and unstructured data.\nIt is used to drive bus
iness decisions, optimize operations, \nand build predictive models'
```

Escape Characters

In [ ]:
```
Backslash is a Escape characters.
```

In [7]:
```
string='Python and \'data\'science'
print(string)
string
```

```
Python and 'data'science
```

Out[7]:
```
"Python and 'data'science"
```

In [11]:
```
ng="Data science  is an \n interdisciplinary field that combines statistics\n machine le
t(string)
```

```
Data science  is an interdisciplinary field that combines statistics machine learning,
programming
```

In [15]:
```
string1="C:\Users\Sai\Pictures\what is AI.webp"
print(string1)
```

```
  Cell In[15], line 1
    string1="C:\Users\Sai\Pictures\what is AI.webp"
                                                   ^
SyntaxError: (unicode error) 'unicodeescape' codec can't decode bytes in position 2-3:
truncated \UXXXXXXXX escape
```

In [16]:
```
string1="C:\\Users\\Sai\\Pictures\\what is AI.webp"
print(string1)
```

```
C:\Users\Sai\Pictures\what is AI.webp
```

Indexing and Slicing in string

In [ ]:
```
Indexing will always start from 0.
```

Positive Indexing

In [ ]:
```
string1="Python"
string1[0]=P
string1[1]=y
string1[2]=t
string1[3]=h
string1[4]=o
string1[5]=n

Indexing is used to access only one character .
len() function used for checking length of string.
```

In [ ]:
```
string=  " P      y      t        h      o       n"
+ve    =    0      1       2       3      4      5
-ve    =  -6      -5      -4      -3     -2     -1
```

In [17]:
```
string="Python"
x=len(string)
print(x)
```

```
6
```

In [20]:
```
string="Python"
x=len(string)
string[x]
```

```
---------------------------------------------------------------------------
IndexError                                Traceback (most recent call last)
Cell In[20], line 3
      1 string="Python"
      2 x=len(string)
----> 3 string[x]

IndexError: string index out of range
```

In [21]:
```
string="Python"
x=len(string)
string[x-1]
```

Out[21]: `'n'`

In [22]:
```
string='python'
string[len(string)-1]
```

Out[22]: `'n'`

In [23]:
```
x='12345'
len(x)
```

Out[23]: `5`

In [24]:
```
x=''
len(x)
```

Out[24]: `0`

Negative Indexing

In [ ]:
```
string1="Python"
string1[-1]=n
string1[-2]=o
string1[-3]=h
string1[-4]=t
string1[-5]=y
string1[-6]=P
```

In [25]:
```
string="Python"
print(string[2])
print(string[-4])
```

```
t
t
```

Slicing

In [ ]:
```
Slicing is used to access one or more chatcters of string.

string[start_index:end_index:[step_size]]

start_index>> include
end_index>> exclude

default start_index =0
default end_index =len(string)
```

In [26]:
```
string="python"
print(string[2])
print(string[3])
print(string[4])
```

```
t
h
o
```

In [ ]:
```
string=  " P     y      t        h      o       n"
+ve   =    0     1       2       3      4      5
-ve   =  -6     -5     -4      -3     -2     -1
```

In [27]:
```
string="python"
string[2:5]#tho
```

Out[27]: 'tho'

In [28]:
```
string="python"
string[1:5]
```

Out[28]: 'ytho'

In [29]:
```
string="machine learning"
string[3:7]
```

Out[29]: 'hine'

In [30]:
```
string="machine learning"
string[3:8]
```
Out[30]: 'hine '

In [32]:
```
string="python"
string[1:4]
string[-5:-2]
```
Out[32]: 'yth'

In [33]:
```
string="python"
string[:]
```
Out[33]: 'python'

In [34]:
```
string="python"
string[0:]
```
Out[34]: 'python'

In [35]:
```
string="python"
string[:len(string)]
```
Out[35]: 'python'

In [ ]:
```
string=  " P     y      t        h     o       n"
+ve   =    0     1       2       3     4      5
-ve   =  -6      -5     -4      -3     -2     -1
```

In [36]:
```
string="python"
print(string[1:4]) #yth
print(string[-5:-2])#yth
print(string[1:-2])#yth
print(string[-5:4])#yth
```
```
yth
yth
yth
yth
```

How to find first 5 character from string

In [41]:
```
string="DataScience and Machine Learning"
string[:5]
```
Out[41]: 'DataS'

How to find last n characters

In [42]:
```
string="DataScience and Machine Learning"
string[-5:]
```
Out[42]: 'rning'

In [43]:
```
string="DataScience and Machine Learning"
string[-10:]
```

Out[43]: 'e Learning'

In [44]:
```
string="DataScience and Machine Learning"
string[-10:-1]
```

Out[44]: 'e Learnin'

In [46]:
```
string="python"
string[1:4]
string[4:1]
```

Out[46]: ''

In [47]:
```
string="Python and Data science"
string[-10:-5] #-10 -9 -8 -7 -6
```

Out[47]: 'ta sc'

Reverse String

In [49]:
```
string1="python"
string1[::-1]
```

Out[49]: 'nohtyp'

In [50]:
```
string="python"
string[5:2:-1]
```

Out[50]: 'noh'

In [51]:
```
string="machine learning"
string[::-2]
```

Out[51]: 'gire nha'

In [55]:
```
string="machine learning"
string[2:8:-2]
```

Out[55]: ''

In [56]:
```
string="machine learning"
string[2:8:2]
```

Out[56]: 'cie'

In [ ]: